inductively, let $A_k \subset C_k$ be the set of cubes disjoint from K and not contained in cubes of A_j for $j < k$. The open set X is then the union of all the cubes in the combined collection $A = \bigcup_k A_k$. Note that the collection A is locally finite: Each point of X has a neighborhood meeting only finitely many cubes in A, since the point has a positive distance from the closed set K.

If two cubes of A intersect, their intersection is an i-dimensional face of one of them for some $i < n$. Likewise, when two faces of cubes of A intersect, their intersection is a face of one of them. This implies that the open faces of cubes of A that are minimal with respect to inclusion among such faces form the cells of a CW structure on X, since the boundary of such a face is a union of such faces. The vertices of this CW structure are thus the vertices of all the cubes of A, and the n-cells are the interiors of the cubes of A.

Next we define inductively a subcomplex Z of this CW structure on X and a map $r : Z \to K$. The 0-cells of Z are exactly the 0-cells of X, and we let r send each 0-cell to the closest point of K, or if this is not unique, any one of the closest points of K. Assume inductively that Z^k and $r : Z^k \to K$ have been defined. For a cell e^{k+1} of X with boundary in Z^k, if the restriction of r to this boundary extends over e^{k+1} then we include e^{k+1} in Z^{k+1} and we let r on e^{k+1} be such an extension that is not too large, say an extension for which the diameter of its image $r(e^{k+1})$ is less than twice the infimum of the diameters for all possible extensions. This defines Z^{k+1} and $r : Z^{k+1} \to K$. At the end of the induction we set $Z = Z^n$.

It remains to verify that by letting r equal the identity on K we obtain a continuous retraction $Z \cup K \to K$, and that $Z \cup K$ contains a neighborhood of K. Given a point $x \in K$, let U be a ball in the metric space K centered at x. Since K is locally contractible, we can choose a finite sequence of balls in K centered at x, of the form $U = U_n \supset V_n \supset U_{n-1} \supset V_{n-1} \supset \cdots \supset U_0 \supset V_0$, each ball having radius equal to some small fraction of the radius of the preceding one, and with V_i contractible in U_i. Let $B \subset \mathbb{R}^n$ be a ball centered at x with radius less than half the radius of V_0, and let Y be the subcomplex of X formed by the cells whose closures are contained in B. Thus $Y \cup K$ contains a neighborhood of x in $\mathbb{R}^n$. By the choice of B and the definition of r on 0-cells we have $r(Y^0) \subset V_0$. Since V_0 is contractible in U_0, r is defined on the 1-cells of Y. Also, $r(Y^1) \subset V_1$ by the definition of r on 1-cells and the fact that U_0 is much smaller than V_1. Similarly, by induction we have r defined on Y^i with $r(Y^i) \subset V_i$ for all i. In particular, r maps Y to U. Since U could be arbitrarily small, this shows that extending r by the identity map on K gives a continuous map $r : Z \cup K \to K$. And since $Y \subset Z$, we see that $Z \cup K$ contains a neighborhood of K by the earlier observation that $Y \cup K$ contains a neighborhood of x. Thus $r : Z \cup K \to K$ retracts a neighborhood of K onto K.

Now for the converse. Since open sets in $\mathbb{R}^n$ are locally contractible, it suffices to show that a retract of a locally contractible space is locally contractible. Let $r : X \to A$

be a retraction and let $U \subset A$ be a neighborhood of a given point $x \in A$. If X is locally contractible, then inside the open set $r^{-1}(U)$ there is a neighborhood V of x that is contractible in $r^{-1}(U)$, say by a homotopy $f_t : V \to r^{-1}(U)$. Then $V \cap A$ is contractible in U via the restriction of the composition rf_t. □

A space X is called a **Euclidean neighborhood retract** or **ENR** if for some n there exists an embedding $i : X \hookrightarrow \mathbb{R}^n$ such that $i(X)$ is a retract of some neighborhood in $\mathbb{R}^n$. The preceding theorem implies that the existence of the retraction is independent of the choice of embedding, at least when X is compact.

Corollary A.8. *A compact space is an ENR iff it can be embedded as a retract of a finite simplicial complex. Hence the homology groups and the fundamental group of a compact ENR are finitely generated.*

Proof: A finite simplicial complex K with n vertices is a subcomplex of a simplex Δ^{n-1}, and hence embeds in $\mathbb{R}^n$. The preceding theorem then implies that K is a retract of some neighborhood in $\mathbb{R}^n$, so any retract of K is also a retract of such a neighborhood, via the composition of the two retractions. Conversely, let K be a compact space that is a retract of some open neighborhood U in $\mathbb{R}^n$. Since K is compact it is bounded, lying in some large simplex $\Delta^n \subset \mathbb{R}^n$. Subdivide Δ^n, say by repeated barycentric subdivision, so that all simplices of the subdivision have diameter less than the distance from K to the complement of U. Then the union of all the simplices in this subdivision that intersect K is a finite simplicial complex that retracts onto K via the restriction of the retraction $U \to K$. □

Corollary A.9. *Every compact manifold, with or without boundary, is an ENR.*

Proof: Manifolds are locally contractible, so it suffices to show that a compact manifold M can be embedded in $\mathbb{R}^k$ for some k. If M is not closed, it embeds in the closed manifold obtained from two copies of M by identifying their boundaries. So it suffices to consider the case that M is closed. By compactness there exist finitely many closed balls $B_i^n \subset M$ whose interiors cover M, where n is the dimension of M. Let $f_i : M \to S^n$ be the quotient map collapsing the complement of the interior of B_i^n to a point. These f_i's are the components of a map $f : M \to (S^n)^m$ which is injective since if x and y are distinct points of M with x in the interior of B_i^n, say, then $f_i(x) \neq f_i(y)$. Composing f with an embedding $(S^n)^m \hookrightarrow \mathbb{R}^k$, for example the product of the standard embeddings $S^n \hookrightarrow \mathbb{R}^{n+1}$, we obtain a continuous injection $M \hookrightarrow \mathbb{R}^k$, and this is a homeomorphism onto its image since M is compact. □

Corollary A.10. *Every finite CW complex is an ENR.*

Proof: Since CW complexes are locally contractible, it suffices to show that a finite CW complex can be embedded in some $\mathbb{R}^n$. This is proved by induction on the number of cells. Suppose the CW complex X is obtained from a subcomplex A by attaching a cell e^k via a map $f:S^{k-1}\to A$, and suppose that we have an embedding $A\hookrightarrow\mathbb{R}^m$. Then we can embed X in $\mathbb{R}^k\times\mathbb{R}^m\times\mathbb{R}$ as the union of $D^k\times\{0\}\times\{0\}$, $\{0\}\times A\times\{1\}$, and all line segments joining points $(x,0,0)$ and $(0,f(x),1)$ for $x\in S^{k-1}$. □

Spaces Dominated by CW Complexes

We have been considering spaces which are retracts of finite simplicial complexes, and now we show that such spaces have the homotopy type of CW complexes. In fact, we can just as easily prove something a little more general than this. A space Y is said to be **dominated** by a space X if there are maps $Y\xrightarrow{i}X\xrightarrow{r}Y$ with $ri\simeq\mathbb{1}$. This makes the notion of a retract into something that depends only on the homotopy types of the spaces involved.

Proposition A.11. *A space dominated by a CW complex is homotopy equivalent to a CW complex.*

Proof: Recall from §3.F that the mapping telescope $T(f_1,f_2,\cdots)$ of a sequence of maps $X_1\xrightarrow{f_1}X_2\xrightarrow{f_2}X_3\longrightarrow\cdots$ is the quotient space of $\coprod_i(X_i\times[i,i+1])$ obtained by identifying $(x,i+1)\in X_i\times[i,i+1]$ with $(f(x),i+1)\in X_{i+1}\times[i+1,i+2]$. We shall need the following elementary facts:

(1) $T(f_1,f_2,\cdots)\simeq T(g_1,g_2,\cdots)$ if $f_i\simeq g_i$ for each i.
(2) $T(f_1,f_2,\cdots)\simeq T(f_2,f_3,\cdots)$.
(3) $T(f_1,f_2,\cdots)\simeq T(f_2f_1,f_4f_3,\cdots)$.

The second of these is obvious. To prove the other two we will use Proposition 0.18, whose proof applies not just to CW pairs but to any pair (X_1,A) for which there is a deformation retraction of $X_1\times I$ onto $X_1\times\{0\}\cup A\times I$. To prove (1) we regard $T(f_1,f_2,\cdots)$ as being obtained from $\coprod_i(X_i\times\{i\})$ by attaching $\coprod_i(X_i\times[i,i+1])$. Then we can obtain $T(g_1,g_2,\cdots)$ by varying the attaching map by homotopy. To prove (3) we view $T(f_1,f_2,\cdots)$ as obtained from the disjoint union of the mapping cylinders $M(f_{2i})$ by attaching $\coprod_i(X_{2i-1}\times[2i-1,2i])$. By sliding the attachment of $X\times[2i-1,2i]$ to $X_{2i}\subset M(f_{2i})$ down the latter mapping cylinder to X_{2i+1} we convert $M(f_{2i-1})\cup M(f_{2i})$ into $M(f_{2i}f_{2i-1})\cup M(f_{2i})$. This last space deformation retracts onto $M(f_{2i}f_{2i-1})$. Doing this for all i gives the homotopy equivalence in (3).

Now to prove the proposition, suppose that the space Y is dominated by the CW complex X via maps $Y\xrightarrow{i}X\xrightarrow{r}Y$ with $ri\simeq\mathbb{1}$. By (2) and (3) we have $T(ir,ir,\cdots)\simeq T(r,i,r,i,\cdots)\simeq T(i,r,i,r,\cdots)\simeq T(ri,ri,\cdots)$. Since $ri\simeq\mathbb{1}$, $T(ri,ri,\cdots)$ is homotopy equivalent to the telescope of the identity maps $Y\to Y\to Y\to\cdots$, which is $Y\times[0,\infty)\simeq Y$. On the other hand, the map ir is homotopic to a cellular map $f:X\to X$, so $T(ir,ir,\cdots)\simeq T(f,f,\cdots)$, which is a CW complex. □

One might ask whether a space dominated by a finite CW complex is homotopy equivalent to a finite CW complex. In the simply-connected case this follows from Proposition 4C.1 since such a space has finitely generated homology groups. But there are counterexamples in the general case; see [Wall 1965].

In view of Corollary A.10 the preceding proposition implies:

Corollary A.12. *A compact manifold is homotopy equivalent to a CW complex.* □

One could ask more refined questions. For example, do all compact manifolds have CW complex structures, or even simplicial complex structures? Answers here are considerably harder to come by. Restricting attention to closed manifolds for simplicity, the present status of these questions is the following. For manifolds of dimensions less than 4, simplicial complex structures always exist. In dimension 4 there are closed manifolds that do not have simplicial complex structures, while the existence of CW structures is an open question. In dimensions greater than 4, CW structures always exist, but whether simplicial structures always exist is unknown, though it is known that there are n-manifolds not having simplicial structures locally isomorphic to any linear simplicial subdivision of $\mathbb{R}^n$, for all $n \geq 4$. For more on these questions, see [Kirby & Siebenmann 1977] and [Freedman & Quinn 1990].

Exercises

1. Show that a covering space of a CW complex is also a CW complex, with cells projecting homeomorphically onto cells.

2. Let X be a CW complex and x_0 any point of X. Construct a new CW complex structure on X having x_0 as a 0-cell, and having each of the original cells a union of the new cells. The latter condition is expressed by saying the new CW structure is a **subdivision** of the old one.

3. Show that a CW complex is path-connected iff its 1-skeleton is path-connected.

4. Show that a CW complex is locally compact iff each point has a neighborhood that meets only finitely many cells.

5. For a space X, show that the identity map $X_c \to X$ induces an isomorphism on π_1, where X_c denotes X with the compactly generated topology.

The Compact-Open Topology

By definition, the compact-open topology on the space X^Y of maps $f: Y \to X$ has a subbasis consisting of the sets $M(K,U)$ of mappings taking a compact set $K \subset Y$ to an open set $U \subset X$. Thus a basis for X^Y consists of sets of maps taking a finite number of compact sets $K_i \subset Y$ to open sets $U_i \subset X$. If Y is compact, which is the only case we consider in this book, convergence to $f \in X^Y$ means, loosely speaking,